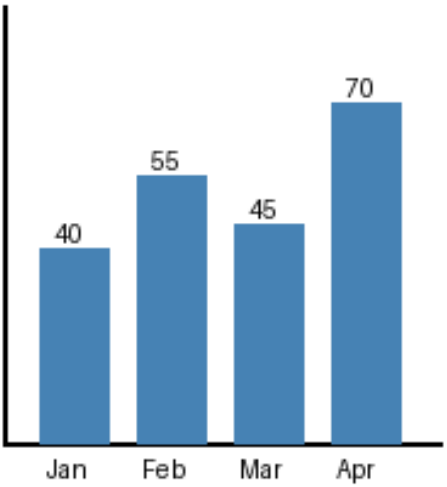
Sample Screening Paper - 20 marks - 60 minutes

Section A: Quantitative Aptitude (2 marks)

Q1. Refer to the bar chart. By what percentage did sales rise from March to April? [1 ma



(a) 25%
(b) 45%
(c) 55.6% (approx.)
(d) 62.5%

Q2. A train 180 m long crosses a pole in 12 seconds. Its speed is? [1 mark]
(a) 10 m/s
(b) 15 m/s
(c) 20 m/s
(d) 25 m/s

Section B: English (1 mark)

Q3. Choose the word closest in meaning to EPHEMERAL. [1 mark]
(a) Eternal
(b) Short-lived
(c) Fragile
(d) Obvious

Section D: Coding (8 marks)

Q4. Balanced Brackets [8 marks]

A string of the characters ( ) [ ] { } is balanced if every opening bracket is closed by the sa
in the correct order. Write a program that tells whether the given string is balanced.

Input format: One line containing the string s.
Output format: Print YES if the string is balanced, otherwise print NO.
Constraints: 1 <= |s| <= 100000.

Sample: {[()]} gives YES, ([)] gives NO.

Section E: SQL (9 marks)

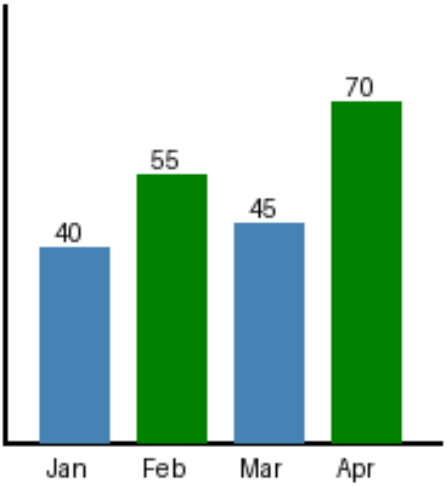
Q5. Employees earning more than 50000 [9 marks]
Write one SQLite query returning the names of employees who earn more than 50000, s
Table employees(id INTEGER PRIMARY KEY, name TEXT, salary INTEGER). Output c

Answer Key and Solutions

For the examiner only. Do not share with candidates.

Q1 (c) Q2 (b) Q3 (b)

Q1. March = 45 and April = 70, so the increase is 25/45 = 55.6%. The green bars mark m



Q4 reference solution (Python):

```
s = input().strip()
pairs = {')': '(', ']': '[', '}': '{'}
st = []
ok = True
for ch in s:
    if ch in '([{':
        st.append(ch)
    elif not st or st.pop() != pairs[ch]:
        ok = False
        break
print('YES' if ok and not st else 'NO')
```

Q4 test cases: 1 open {[()]} -> YES; 2 open ([)] -> NO; 3 hidden ()[]{} -> YES; 4 hidden
5 hidden maximum size, deeply nested -> YES.

Q5 reference query: SELECT name FROM employees WHERE salary > 50000 ORDER
Sample data: INSERT INTO employees VALUES (1,'Asha',60000),(2,'Ben',40000),(3,'Ch
Hidden data: INSERT INTO employees VALUES (1,'Zed',50000),(2,'Amy',50001);